

SAVITRIBAI PHULE PUNE UNIVERSITY

A PRELIMINARY PROJECT REPORT ON

NEXT GENERATION ANALYTICS FOR BNA

**SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS OF**

BACHELOR OF ENGINEERING (Computer Engineering)

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CERTIFICATE

This is to certify that the Project Entitled

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is a bonafide work carried out by Students under the supervision of Prof. M.S. Takalikar and it is submitted towards the partial fulfillment of the requirement of Bachelor of Engineering (Computer Engineering) Project.

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Abstract

BMC BladeLogic Network Automation (BNA) is a network automation and management software for managing network devices within the data centres of an enterprise spread across the globe. It reduces network outages and downtime by automating configuration changes and compliance processes. Our project aims to build an advanced data analytics and reporting platform that explores the extensive BNA data set utilizing state-of-the-art machine learning algorithms. The data mining tool will be designed with a modular, concurrent and scalable architecture using JAVA and JVM-based toolkits such as Akka. Offered as a RESTful web service, it will support various data mining techniques (clustering, regression, classification and association), predictive analytics, statistical modelling and evaluation. To make it more powerful learning schemes, models and algorithms from WEKA and R scripts customized according to the requirements of BNA will be implemented. Additionally, using Yellowfin BI Platform we will introduce dashboards for live reporting. The proposed software system will help gain meaningful insights, ease decision making, make possible adhoc and sophisticated reporting to substitute earlier static reports devoid of applied analytics.

Acknowledgments

*It gives us great pleasure in presenting the preliminary project report on ‘**Next Generation Analytics for BNA**’.*

*We would like to take this opportunity to thank my internal guide **Prof. M.S. Takalikar** for giving me all the help and guidance we needed. We are really grateful to them for their kind support. Their valuable suggestions were very helpful.*

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*In the end our special thanks to **Mr. Jatin Parikh and Mr. Makrand Rahane** for providing various resources such as laboratory with all needed software platforms, continuous Internet connection, for Our Project.*

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CHAPTER 1

SYNOPSIS

1.1 PROJECT TITLE

NEXT GENERATION ANALYTICS OF BNA

1.2 PROJECT OPTION

INDUSTRY SPONSORED

1.3 INTERNAL GUIDE

Prof. M.S. TAKALIKAR

1.4 SPONSORSHIP AND EXTERNAL GUIDE

Mr. JATIN PARIKH Mr. MAKRAND RAHANE (BMC SOFTWARE)

1.5 TECHNICAL KEYWORDS (AS PER ACM KEYWORDS)

1. DATA MINING
2. BUSINESS INTELLIGENCE
3. DATA WAREHOUSING
4. NETWORK SECURITY
5. MACHINE LEARNING

1.6 PROBLEM STATEMENT

Building an advanced data analytics and reporting platform with modular, concurrent and scalable architecture data mining tool for displaying customized BMC Network Automation reports using Yellowfin BI.

1.7 ABSTRACT

BMC BladeLogic Network Automation (BNA) is a network automation and management software for managing network devices within the data centres of an enterprise spread across the globe. It reduces network outages and downtime by automating configuration changes and compliance processes. Our project aims to build an advanced data analytics and reporting platform that explores the extensive BNA data set utilizing state-of-the-art machine learning algorithms. The data mining tool will be designed with a modular, concurrent and scalable architecture using JAVA and JVM-based toolkits such as Akka. Offered as a RESTful web service, it will support various data mining techniques (clustering, regression, classification and association), predictive analytics, statistical modelling and evaluation. To make it more powerful learning schemes, models and algorithms from WEKA and R scripts customized according to the requirements of BNA will be implemented. Additionally, using Yellowfin BI Platform we will introduce dashboards for live reporting. The proposed software system will help gain meaningful insights, ease decision making, make possible adhoc and sophisticated reporting to substitute earlier static reports devoid of applied analytics.

1.8 GOALS AND OBJECTIVES

- Constructing XML file of input, processing logic and output
- Making provisions for RESTful web service
- Deciding data mining algorithms to be implemented
- Deriving Reporting Intelligent Insights for BMC Network Automation by leveraging technologies like:- Data Mining, Business Intelligence, Machine Learning, Knowledge management systems.
- Developing Dynamic Reports.
- Developing Live Reports/Dashboards.

- Provide Adhoc Reporting Capabilities.

1.9 MATHEMATICAL MODEL :

Let S be the system to implement a lexical analyzer.

$$S = \{s, e, X, Y, F_{me}, DD, NDD\}$$

where,

s = start state $\{fl\}$

where,

fl = { setting up database in PostgreSQL , configuring YellowFin }

e = end state { yellowfin reports displayed }

X = set of inputs.

$$= \{X_p, X_t, X_s\}$$

where

X_p = Required reports

$= \{X_p \in \{ \text{Dashboard, Job Summary, Quarantine, Device inventory, OS image history} \} \}$

X_t = Knowledge modules

$$= \{ \langle l_i, t_i, s_i \rangle \mid i \in (1, n) \}$$

where,

n = number of use cases.

l_i = input

t_i = logic

s_i = output

O_s = { Requested reports }

F_{me} = Set of functions.

$$= \{f_1, f_2, f_3\}$$

where,

$f_1 = f_l \rightarrow O_p$
= { accepts input from user and generates XML file }

$f_2 = O_p \rightarrow O_t$
= { establishes connection of database }

$f_3 = O_t \rightarrow O_s$
= { parses XML file }

$f_4 = O_t \rightarrow O_s$
= { reserves temporary table for output generation }

$f_5 = O_t \rightarrow O_s$
= { displays yellowfin reports on UI }

DD = Deterministic Data.
= { set of tables in database }

NDD = Non-Deterministic Data.
= { database size }

1.10 NAMES OF CONFERENCES / JOURNALS WHERE PAPERS CAN BE PUBLISHED

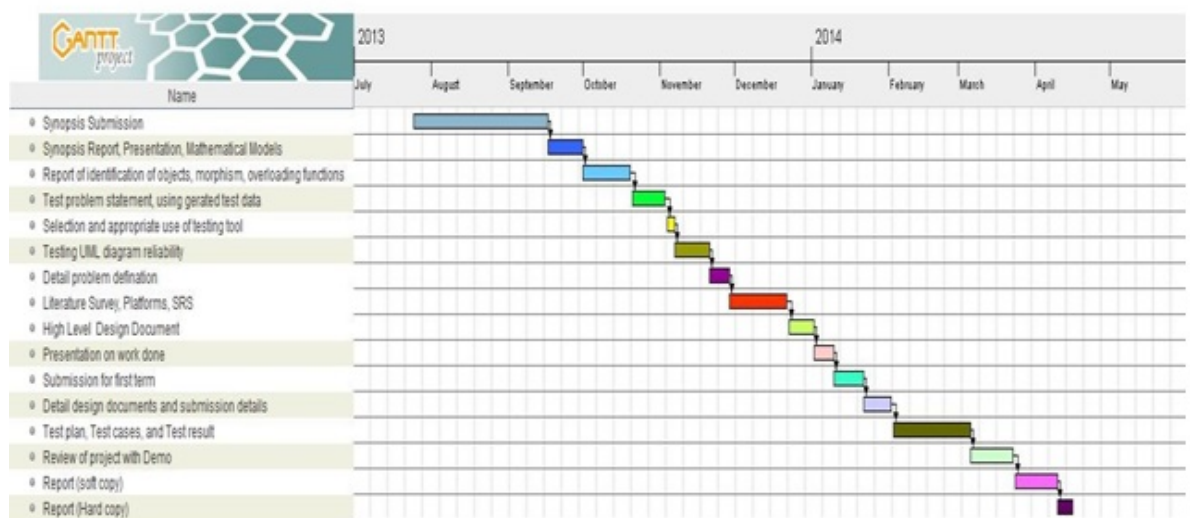
- IJDKP 2016 - International Journal of Data Mining Knowledge Management Process
- ACSTY 2016 Second International Conference on Advances in Computer Science and Information Technology

1.11 REVIEW OF CONFERENCE/JOURNAL PAPERS SUPPORTING PROJECT IDEA

Submitted a paper on "Overview of Data Mining tools for BNA DataSet" to ICEI 2017.

Awaiting reply.

1.12 PLAN OF PROJECT EXECUTION



CHAPTER 2

TECHNICAL KEYWORDS

2.1 AREA OF PROJECT

Computer Networks

2.2 TECHNICAL KEYWORDS

1. DATA MINING
2. BUSINESS INTELLIGENCE
3. DATA WAREHOUSING
4. NETWORK SECURITY
5. MACHINE LEARNING

CHAPTER 3

INTRODUCTION

3.1 PROJECT IDEA

To develop an advanced data analytics and reporting platform which has a modular, concurrent and scalable architecture. A data mining tool for displaying customized BMC Network Automation reports using Yellowfin BI.

3.2 MOTIVATION OF THE PROJECT

SecOps is the seamless collaboration between IT Security and IT Operations to effectively mitigate risk. BMC SecOps solutions enables teams to prioritize and remediate critical vulnerabilities, and systematically address compliance violations through an integrated and automated approach. SecOps solutions for BladeLogic Network Automation help to cope with the volume and complexity of security risks by prioritizing and accelerating closure of known vulnerabilities for networking. They also provide visibility to unmanaged assets which represent potential blindspots in the overall security strategy. BNA natively performs scan-less and real-time detection of vulnerabilities across a diverse network environment without degrading performance. Our data mining tool aims to enable collaboration between security and operations teams for integrated and holistic visibility into risks, impacts, and operational plans. Sophisticated and dynamic reports generated will help in assessing change impact and real-time policy based audits.

3.3 LITERATURE SURVEY

- **BNA:- BMC Network Automation**

BMC Network Automation is the industry-leading solution for automated management, control, and enforcement of network configuration changes in the datacentre .It provides a cross-platform solution for managing both physical and virtual assets in the traditional datacentre and in the Cloud.

- **Yellowfin: Business Intelligence tool**

Yellowfin is a business intelligence (BI), dashboard, reporting and data analysis software vendor. Yellowfins software allows reporting from data stored in relational databases, multi-dimensional cubes or in-memory analytical databases.

- PostgreSQL:-

PostgreSQL, often simply Postgres, is an object-relational database management system (ORDBMS) with an emphasis on extensibility and standards-compliance. As a database server, its primary function is to store data securely, and to allow for retrieval at the request of other software applications. It can handle workloads ranging from small single-machine applications to large Internet-facing applications with many concurrent users.

- RapidMiner:-

Written in the Java Programming language, this tool offers advanced analytics through template-based frameworks. A bonus: Users hardly have to write any code. Offered as a service, rather than a piece of local software, this tool holds top position on the list of data mining tools.

CHAPTER 4

PROBLEM DEFINITION AND SCOPE

4.1 PROBLEM STATEMENT

Building an advanced data analytics and reporting platform with modular, concurrent and scalable architecture for displaying customized BMC Network Automation reports using Yellowfin BI.

4.1.1 Goals and objectives

- Constructing XML file of input, processing logic and output
- Making provisions for RESTful web service
- Deciding data mining algorithms to be implemented
- Deriving Reporting Intelligent Insights for BMC Network Automation by leveraging technologies like:- Data Mining, Business Intelligence ,Machine Learning, Knowledge management systems.
- Developing Dynamic Reports.
- Developing Live Reports/Dashboards.
- Provide Adhoc Reporting Capabilities.

4.1.2 Statement of scope

BNA working on network devices within the data centres of enterprises spreads across the globe generates extensive data .This data is stored in operational databases (Relational) .

4.2 SOFTWARE CONTEXT

Upon successful completion the product it can be embedded in BNA along with YellowFin Reports.

4.3 MAJOR CONSTRAINTS

- 1.Database Discrepancies
- 2.Connection to database

4.4 METHODOLOGIES OF PROBLEM SOLVING AND EFFICIENCY ISSUES

The single problem can be solved by different solutions. This considers the performance parameters for each approach. Thus considers the efficiency issues.

4.5 SCENARIOS IN WHICH MULTI-CORE, EMBEDDED AND DISTRIBUTED COMPUTING USED

- Consider a distributed system with many clients requesting the server through web services, the server, using multi-programming accesses all the requests.
- In very large database sites, multi processing machines are needed for data mining projects; the end user defines the records and attributes to be worked on and an extract from the original database is copied to a multi-processing machine. parallel algorithms for building decision-tree classifiers on shared-memory multiprocessor (SMP) systems could be used.
- Distributed Scenario can be considered when multiple cloud servers are maintained for large network. As future enhancement is considered, distributed computing can be used and the application can be deployed on various servers around the world.

4.6 OUTCOME

Dynamic live reports of existing BNA static reports will be generated using YellowFin with intelligent insights developed on the data mined by the RapidMiner.

4.7 APPLICATIONS

- Intrusion detection
- Research analysis
- Bio Informatics
- Market Basket analysis

4.8 HARDWARE RESOURCES REQUIRED

Sr. No.	Parameter	Minimum Requirement	Justification
1	CPU Speed	2 GHz	Optimum level
2	RAM	8 GB	Sufficient

Table 4.1: Hardware Requirements

4.9 SOFTWARE RESOURCES REQUIRED

Platform :

- Java
- XML (Extensible Mark up Language)
- Apache Tomcat and Axis2
- SAX/DOM XML parsers
- Hibernate
- Postregre SQL

- AKKA
- Yellowfin

CHAPTER 5

PROJECT PLAN

5.1 PROJECT ESTIMATES

5.1.1 Reconciled Estimates

5.1.1.1 Cost Estimate

YellowFin license(bearred by company)

5.1.1.2 Time Estimates

March 2017

5.1.2 Project Resources

- BNA
- PostgreSQL
- YellowFin
- Eclipse IDE

5.2 RISK MANAGEMENT W.R.T. NP HARD ANALYSIS

The data analytics algorithms which we are going to use are challenging and cannot be solved in polynomial time(p). Hence, it is a NP-complete type problem. In computational complexity theory, a decision problem is NP-complete when it is both in NP and NP-hard. The set of NP-complete problems is often denoted by NP-C or NPC. The abbreviation NP refers to "nondeterministic polynomial time".

5.2.1 Risk Identification

- Risk1- Database being centered can crash.
- Risk2- Discrepancies in the database.
- Risk3- server fails to access the request from the client.
- Risk4- lacking job scheduling and servers in the pool remaining idle

5.2.2 Risk Analysis

The risks for the Project can be analyzed within the constraints of time and quality

ID	Risk Description	Probability	Impact		
			Schedule	Quality	Overall
1	Risk1	High	High	High	High
2	Risk2	Low	Low	Low	Low
3	Risk3	Low	High	Medium	Medium
4	Risk4	Low	High	High	High

Table 5.1: Risk Table

Probability	Value	Description
High	Probability of occurrence is	> 75%
Medium	Probability of occurrence is	26 – 75%
Low	Probability of occurrence is	< 25%

Table 5.2: Risk Probability definitions

Impact	Value	Description
Very high	> 10%	Schedule impact or Unacceptable quality
High	5 – 10%	Schedule impact or Some parts of the project have low quality
Medium	< 5%	Schedule impact or Barely noticeable degradation in quality Low Impact on schedule or Quality can be incorporated

Table 5.3: Risk Impact definitions

5.2.3 Overview of Risk Mitigation, Monitoring, Management

Following are the details for each risk.

Risk ID	1
Risk Description	Database being centered can crash
Category	Development Environment.
Source	Database
Probability	High
Impact	High
Response	Monitoring
Strategy	Strategy
Risk Status	Identified

Risk ID	2
Risk Description	database may contain some discrepancies
Category	Requirements
Source	Database
Probability	Low
Impact	Low
Response	Mitigate
Strategy	Better testing will resolve this issue.
Risk Status	Identified

Risk ID	3
Risk Description	server fails to accept client's request
Category	Development Environment
Source	Post-deployment
Probability	Low
Impact	High
Response	Accept
Strategy	search for the authentication.
Risk Status	Identified

Risk ID	4
Risk Description	improper scheduling and servers stay idle in the pool
Category	Development Environment
Source	post-deployment
Probability	Low
Impact	Very High
Response	Mitigate
Strategy	do proper scheduling
Risk Status	Identified

5.3 PROJECT SCHEDULE

5.3.1 Project task set

Major Tasks in the Project stages are:

- Task 1: Studying YellowFin BI tool
- Task 2: Replicating the reports of BNA in YellowFin.
- Task 3: XML file for import-export
- Task 4: making zip files of the Knowledge module and the report for a particular insight
- Task 5: creating a web service scheduling the requests

5.3.2 Task network

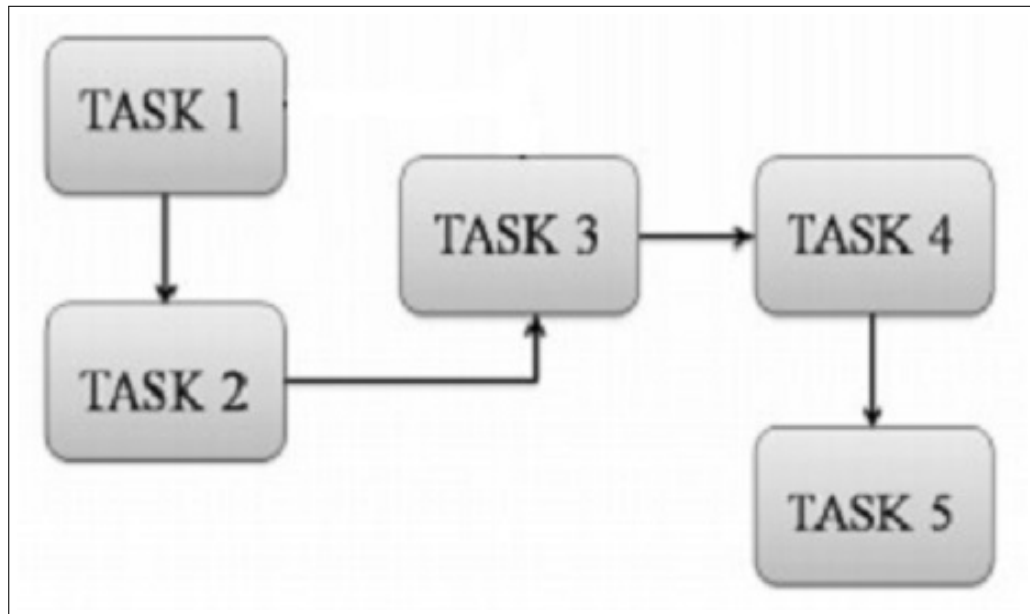


Figure 5.1: Task diagram

5.4 TEAM ORGANIZATION

The manner in which staff is organized and the mechanisms for reporting are noted.

5.4.1 Management reporting and communication

- Email
- Sprint
- Whatsapp
- Verbal Communication

CHAPTER 6

**SOFTWARE REQUIREMENT
SPECIFICATION (SRS IS TO BE
PREPARED USING RELEVANT
MATHEMATICS DERIVED AND
SOFTWARE ENGG. INDICATORS IN
ANNEX A AND B)**

6.1 INTRODUCTION

6.1.1 Purpose and Scope of Document

SecOps is the seamless collaboration between IT Security and IT Operations to effectively mitigate risk. BMC SecOps solutions enables teams to prioritize and remediate critical vulnerabilities, and systematically address compliance violations through an integrated and automated approach. SecOps solutions for BladeLogic Network Automation help to cope with the volume and complexity of security risks by prioritizing and accelerating closure of known vulnerabilities for networking. They also provide visibility to unmanaged assets which represent potential blindspots in the overall security strategy. BNA natively performs scan-less and real-time detection of vulnerabilities across a diverse network environment without degrading performance. Our data mining tool aims to enable collaboration between security and operations teams for integrated and holistic visibility into risks, impacts, and operational plans. Sophisticated and dynamic reports generated will help in assessing change impact and real-time policy based audits.

6.1.2 Overview of responsibilities of Developer

- Deriving Reporting Intelligent Insights for BMC Network Automation by leveraging technologies like:- Data Mining, Business Intelligence , Data warehousing, Big Data, Network Security , Vulnerability , Assessment, Artificial Intelligence, Machine Learning, Knowledge management systems.
- Developing Dynamic Reports.
- Developing Live Reports/Dashboards.
- Provide Adhoc Reporting Capabilities.

6.2 USAGE SCENARIO

This section provides various usage scenarios for the system to be developed.

6.2.1 User profiles

The profiles of all user categories are described here.(Actors and their Description)

6.2.2 Use-cases

All use-cases for the software are presented. Description of all main Use cases using use case template is to be provided.

Use Case	Description	Actors	Assumption
Receive required report	sender sends	sender and client	Request scheduled properly
New table created	table created	client	Table does not exist

Table 6.1: Use case Table

6.2.3 Use Case View

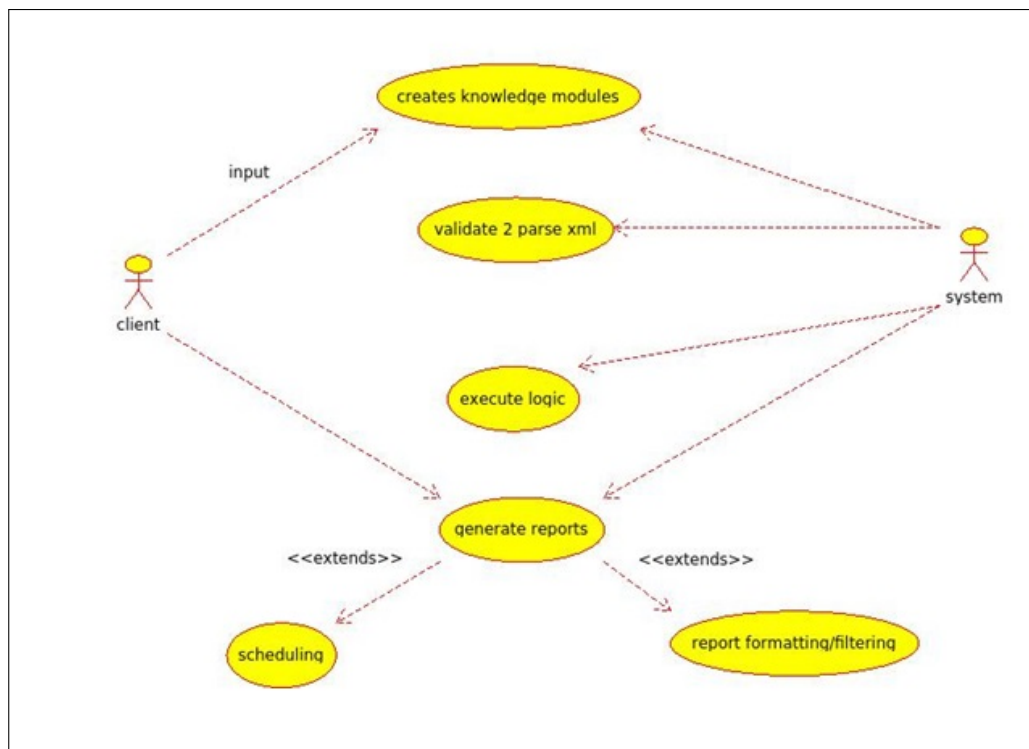


Figure 6.1: Use case diagram

6.3 DATA MODEL AND DESCRIPTION

6.3.1 Data Description

The following kinds of data will be an input to the system:

- .KM zip file of XML file.
- .KM zip file of the report.
- web service.

The data that will be manipulated by the system is:

- KM modules will be unzipped.
- XML files will run.
- scheduling of the requests.

6.3.2 Data objects and Relationships

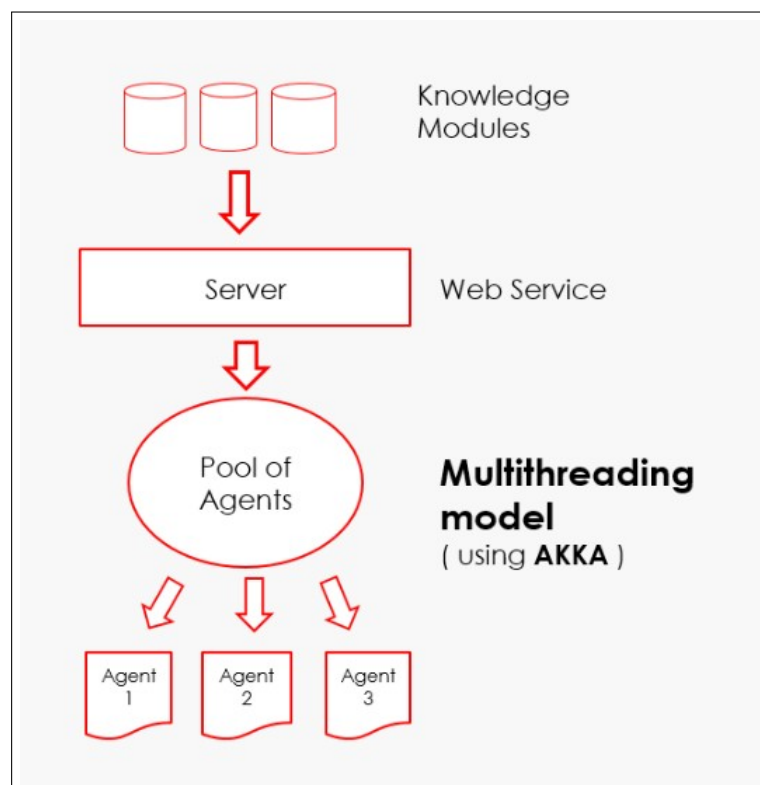


Figure 6.2: Data objects and Relationships

6.4 FUNCTIONAL MODEL AND DESCRIPTION

A description of each major software function, along with data flow (structured analysis) or class hierarchy (Analysis Class diagram with class description for object oriented system) is presented.

6.4.1 Data Flow Diagram

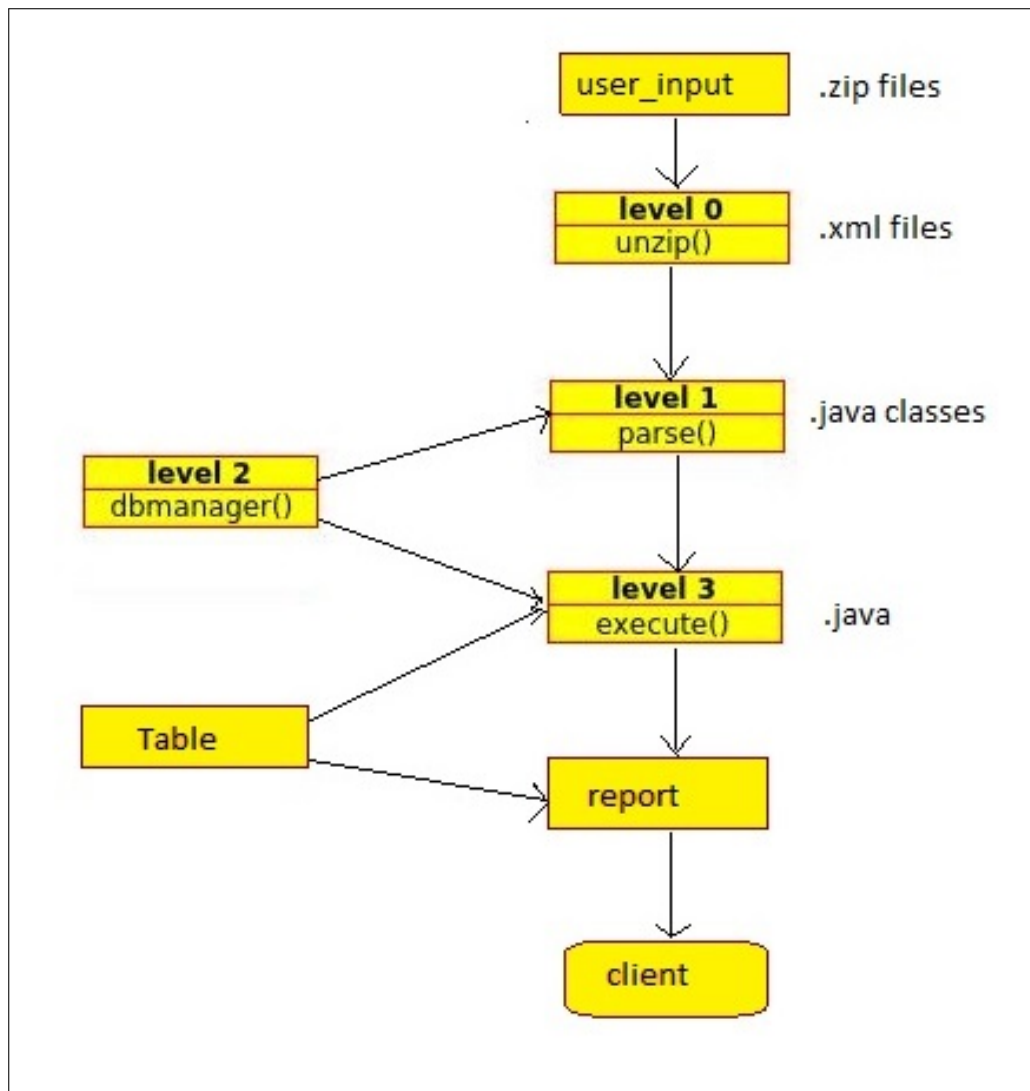


Figure 6.3: Dataflow diagram

6.4.2 Description of functions

Detailed description of the functions:

- Using a XML file, the client can specify which database server he wants to connect to, what algorithm he wants to implement and the output schema he expects. This guarantees a generic framework.
- Using web services, to expose functionality over the network and making messaging more efficient and less bandwidth costly.
- Analyzing the BNA operational databases by applying data mining techniques, will help gain more insights and ease decision making.
- Live reports and dashboards developed using Yellowfin will help assess, change impact and real time policies

6.4.3 Activity Diagram:

- The Activity diagram represents the steps taken.

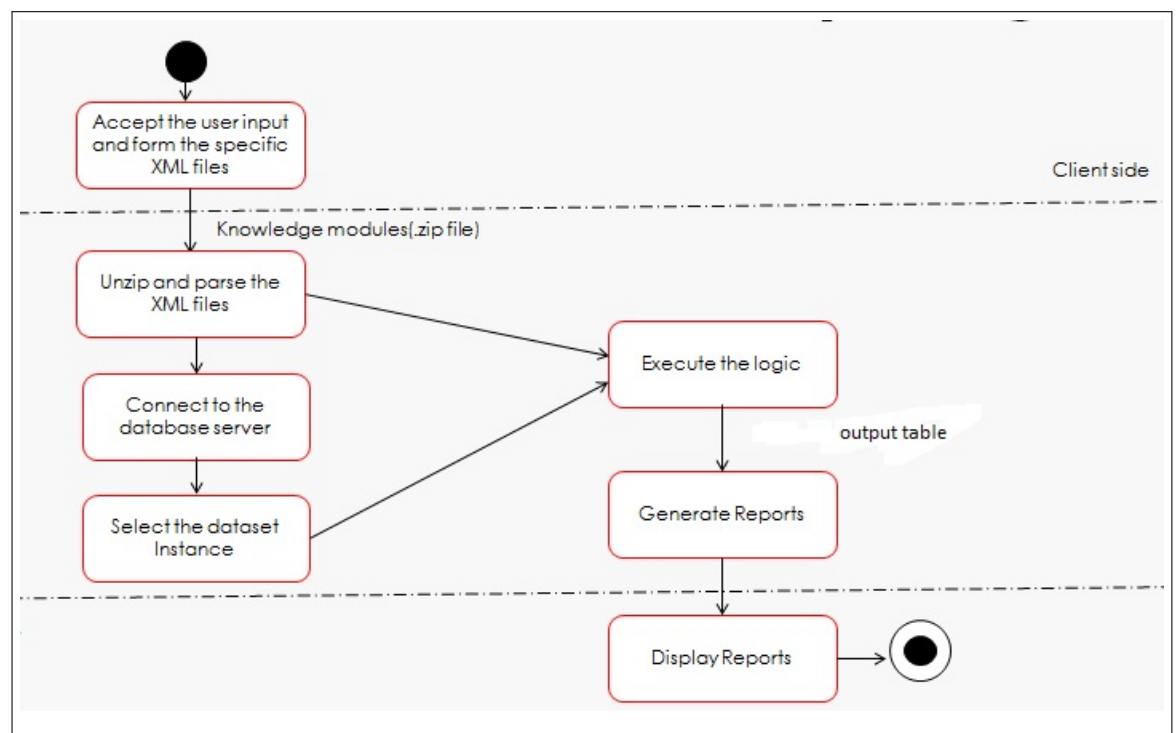


Figure 6.4: Activity diagram

6.4.4 Non Functional Requirements:

- Interface Requirements

- Performance Requirements
- Software quality attributes such as availability [related to Reliability], modifiability [includes portability, reusability, scalability] , performance, security, testability and usability[includes self adaptability and user adaptability]

6.4.5 State Diagram:

State Transition Diagram

The states are represented in ovals and state of system gets changed when certain events occur. The transitions from one state to the other are represented by arrows. The Figure shows important states and events that occur while creating new project.

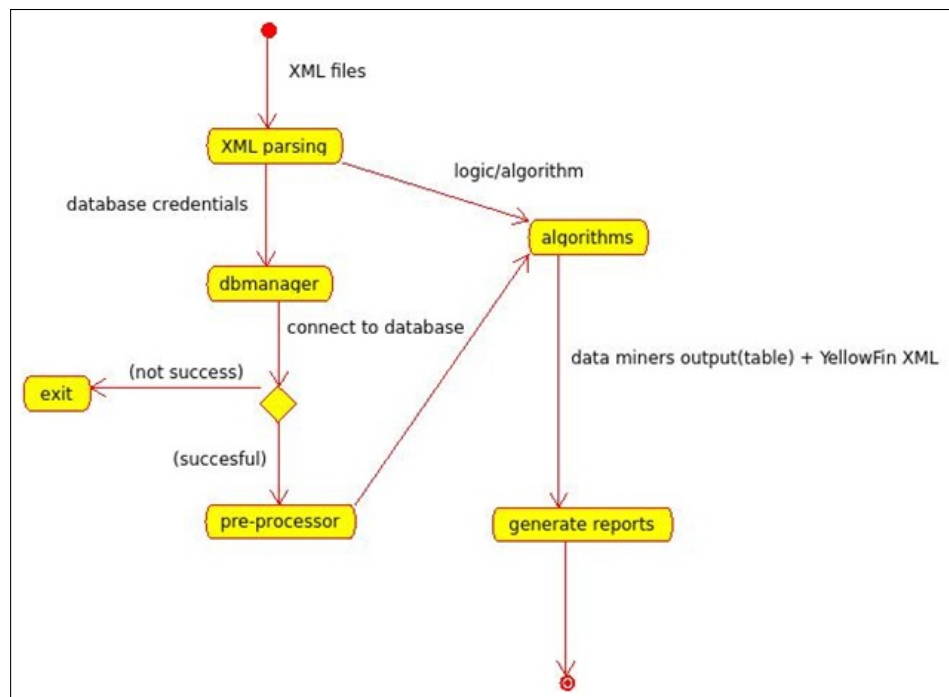


Figure 6.5: State transition diagram

6.4.6 Design Constraints

1)input to the system is in the form of XML file.

6.4.7 Software Interface Description

The software consists of data inputs such as:

- 1) XML file.
- 2) report generated in YellowFin.

CHAPTER 7

DETAILED DESIGN DOCUMENT USING

APPENDIX A AND B

7.1 INTRODUCTION

This document specifies the design that is used to fulfill the aim of Next Gen Analytics for BNA.

7.2 ARCHITECTURAL DESIGN

A description of the program architecture is presented. Subsystem design or Block diagram, Package Diagram, Deployment diagram with description is to be presented.

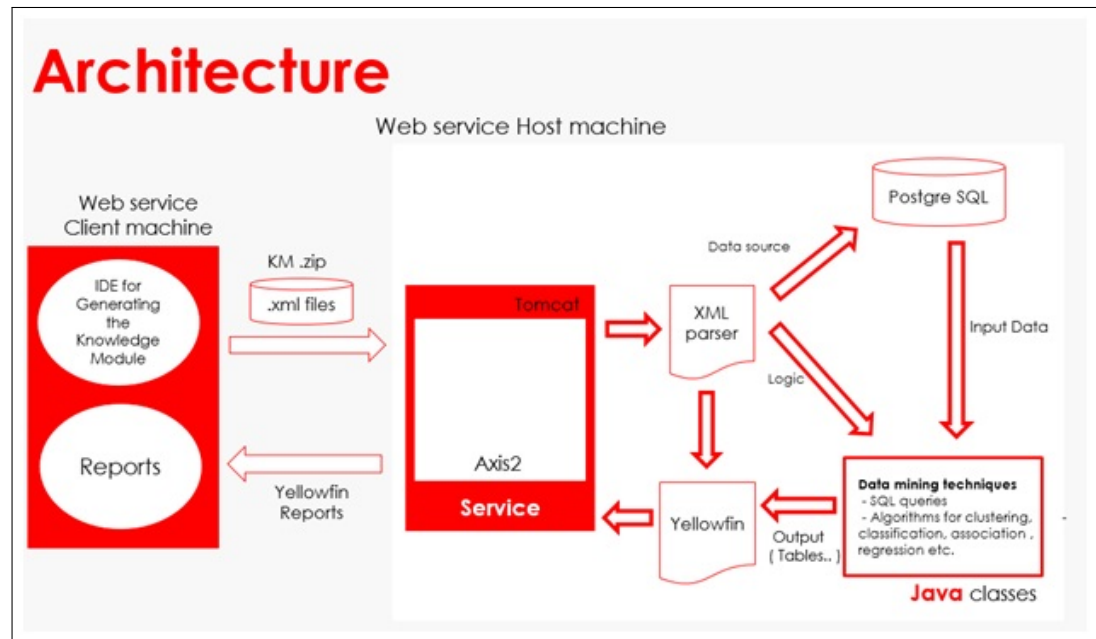


Figure 7.1: Architecture diagram

7.3 DATA DESIGN (USING APPENDICES A AND B)

A description of all data structures including internal, global, and temporary data structures, database design (tables), file formats.

7.3.1 Internal software data structure

- The temporary extracted data is stored in buffers or cache.
- The output is stored in the table given by the SQL query.

7.3.2 Global data structure

Report generated is in the form of a table.

7.3.3 Temporary data structure

Buffer or cache is used for temporary storage.

7.3.4 Database description

Database of 490 tables is accessed using JDBC connection for PostgreSQL.

7.4 COMPONENT DESIGN

Class diagrams, Interaction Diagrams, Algorithms. Description of each component description required.

7.4.1 Class Diagram

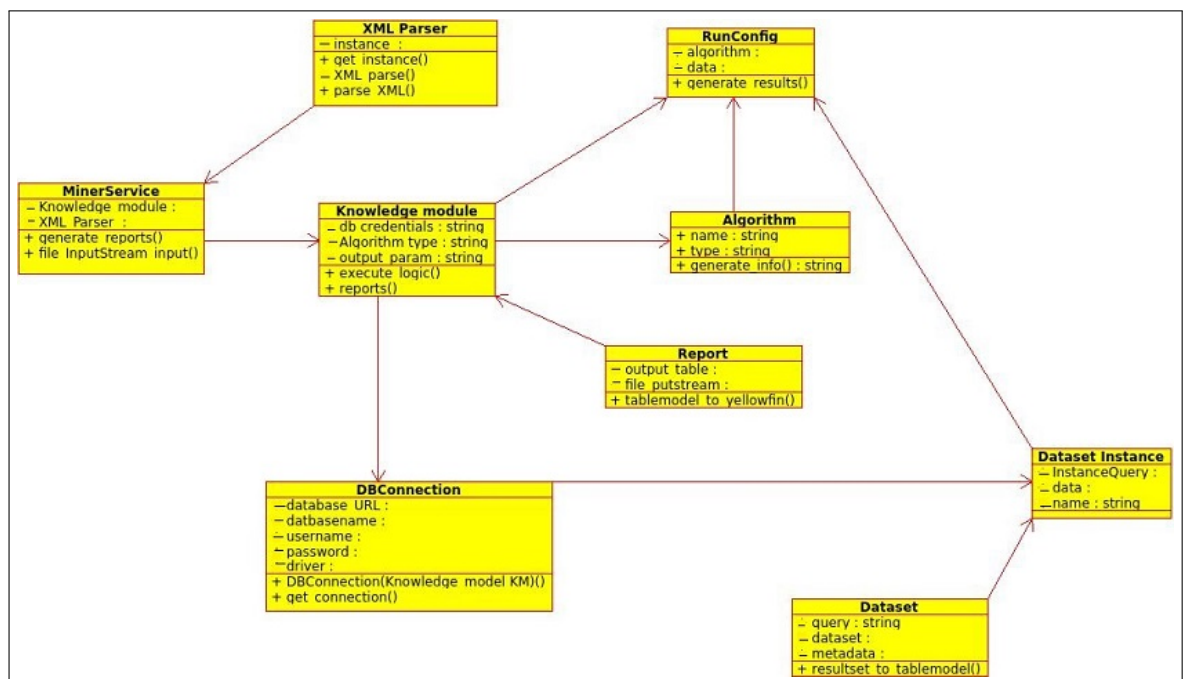


Figure 7.2: Class Diagram

CHAPTER 8

SUMMARY AND CONCLUSION

BMC BladeLogic Network Automation (BNA) is a network automation and management software for managing network devices within the data centres of an enterprise spread across the globe. It reduces network outages and downtime by automating configuration changes and compliance processes. BNA originally consisted of static reports.

Using YellowFin our system developed dynamic and live reports which could be easily embedded in our product and provided to the client when requested.

Our system is an advanced data analytics and reporting platform that explores the extensive BNA data set utilizing state-of-the-art machine learning algorithms. The data mining tool we have designed is a modular, concurrent and scalable architecture developed using JAVA and JVM-based toolkits such as Akka. Offered as a RESTful web service, it supports various data mining techniques (clustering, regression, classification and association), predictive analytics, statistical modelling and evaluation. Additionally, using Yellowfin BI Platform we have introduced dashboards for live reporting.

The proposed software system will help gain meaningful insights, ease decision making, make possible adhoc and sophisticated reporting to substitute earlier static reports devoid of applied analytics.

CHAPTER 9

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ANNEXURE A

LABORATORY ASSIGNMENTS ON

PROJECT ANALYSIS OF ALGORITHMIC

DESIGN

I	D	E	A
Input of 2 xml files	Dynamic reports	Efficient reports	Adhoc reporting
Improved performance	Develop live reports	Eliminate static views	Accessing using UI
Incorporate KM	Deriving insights	Eliminate error	Accelerate o/p

Table A.1: IDEA Matrix

- To develop the problem under consideration and justify feasibility using concepts of knowledge canvas and IDEA Matrix.
- Project problem statement feasibility assessment using NP-Hard, NP-Complete or satisfy ability issues using modern algebra and/or relevant mathematical models.
- input x , output y , $y=f(x)$ The data analytics algorithms which we are going to use are challenging and cannot be solved in polynomial time(p). Hence, it is a NP-complete type problem.

In computational complexity theory, a decision problem is NP-complete when it is both in NP and NP-hard. The set of NP-complete problems is often denoted by NP-C or NPC. The abbreviation NP refers to "nondeterministic polynomial time".

ANNEXURE B

LABORATORY ASSIGNMENTS ON

PROJECT QUALITY AND RELIABILITY

TESTING OF PROJECT DESIGN

- Black Box Testing

Black-box testing is a method of software testing that examines the functionality of an application without peering into its internal structures or workings. This method of test can be applied to virtually every level of software testing: Unit, Integration, System and acceptance. Entire independent modules knowledge module parsing , execution and report generation would be tested.

- White box Testing

White box Testing (known as glass box testing), is used for testing the loops and conditions of the program. It focuses on correct execution rather than fulfilling of requirements. It ensures that each and every statement is executed atleast once. There are two types of white box testing:

1. Basis Path Testing
2. Control Structure Testing

White box testing is verification and validation technique which we have used to examine whether our code works as expected.

- Manual Testing

We have provided appropriate inputs to conform that all modules of our code provide output as expected.

- Risks Handled

- 1)Failure in database connection.
- 2)Discrepancies in database.

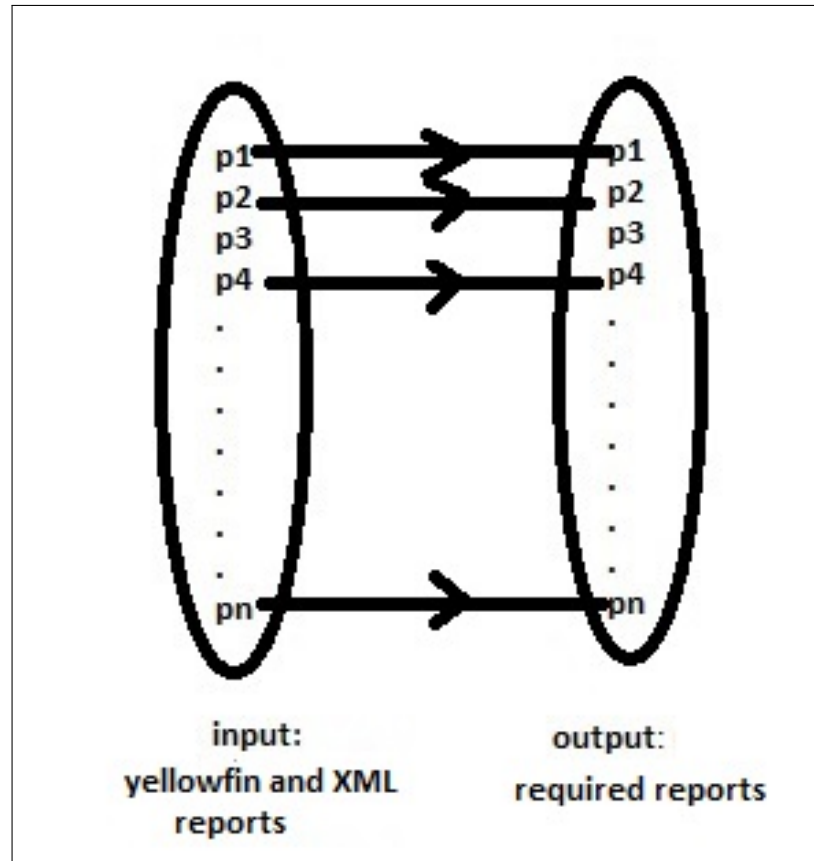


Figure B.1: Venn Diagram

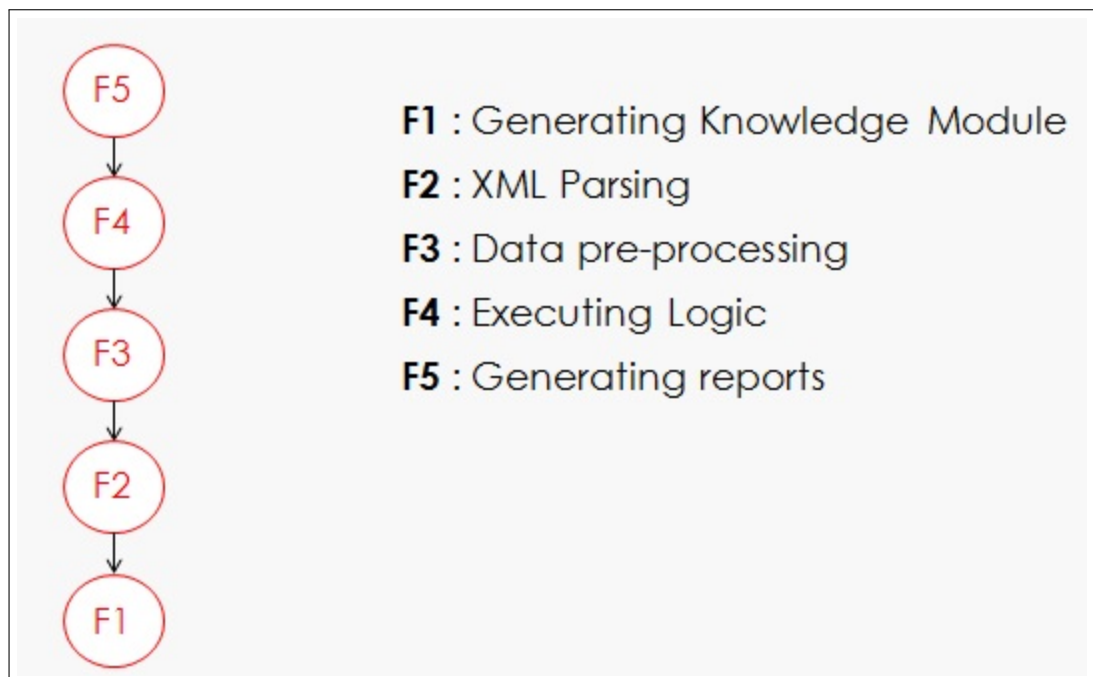


Figure B.2: Function Dependency Diagram

ANNEXURE C

PROJECT PLANNER

Using planner or alike project management tool.

Major Tasks in the Project stages are:

T1: Deciding the boundaries and scope of the Project.

T2: Deciding Inputs and Outputs

T3: Design and Modeling of project

T4: Applying project knowledge and construct a research paper

T5: Installing required software and tools

T6: learning the tools which are required for project

T7: Preparing of prototype

T8: Revision of prototype

T9: Testing

T10: Documentation

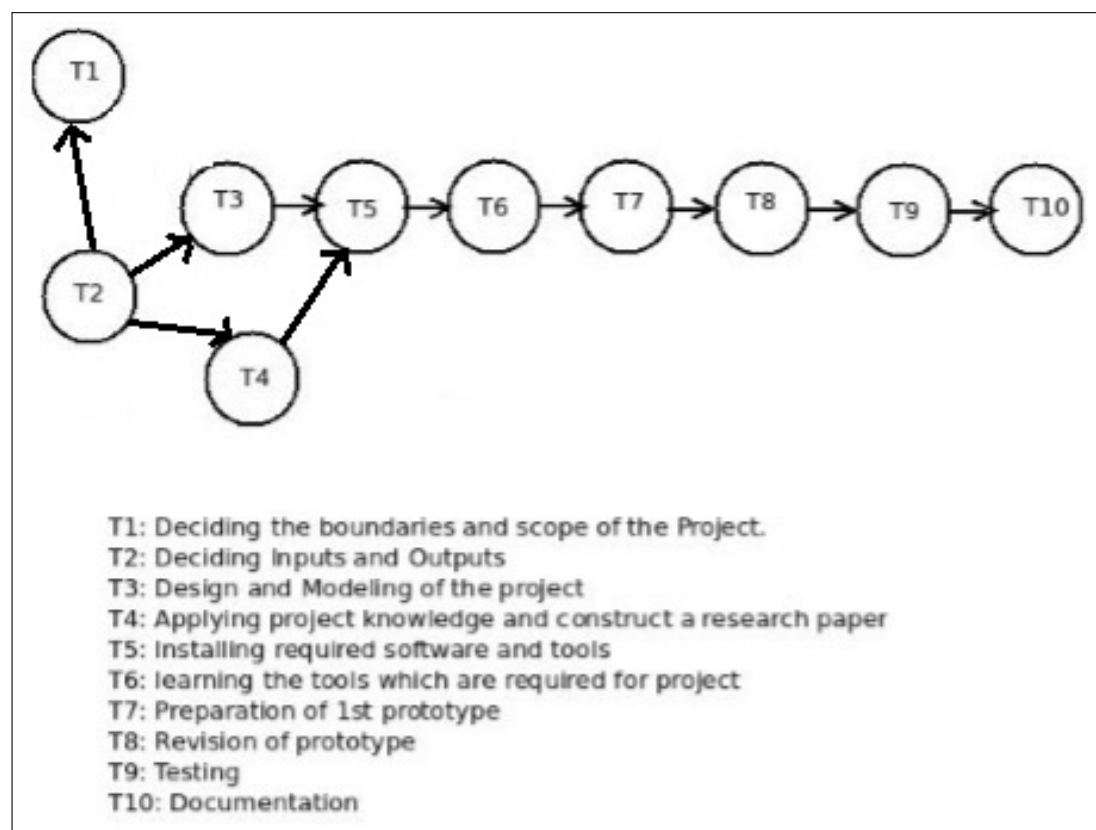


Figure C.1: project planner

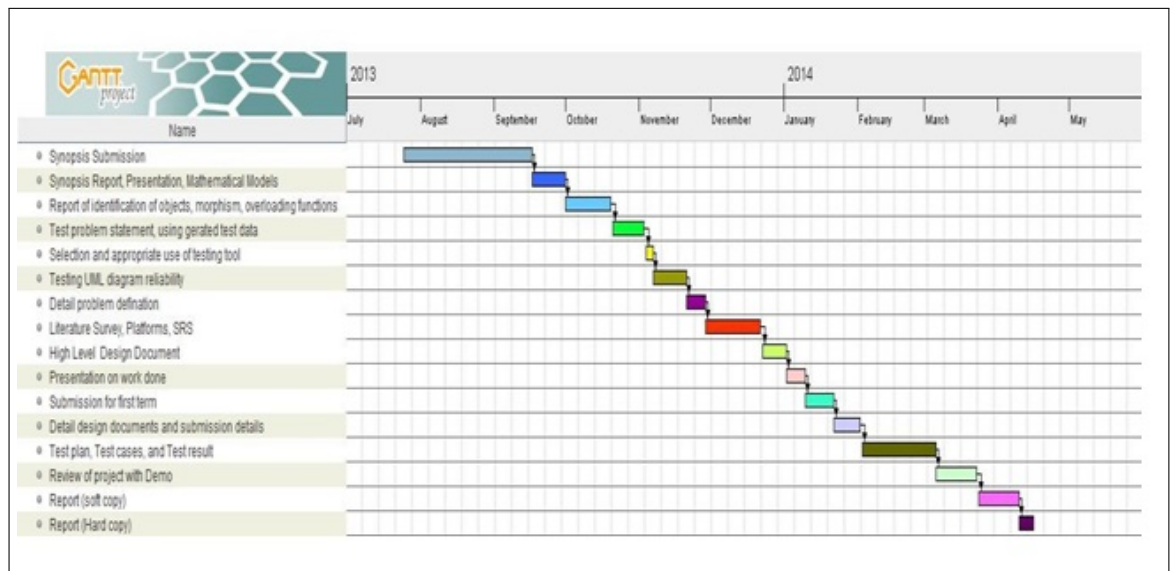


Figure C.2: project planner

ANNEXURE D

PLAGIARISM REPORT